Cause
or Coincidence

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Watch: Cause or Coincidence?

- Full narrated walkthrough of the ideas covered in this deck
- Cause or Coincidence?

1. The Core Challenge

> *"To understand cause and effect, you can't just sit back and watch; you have to interfere."*

- **Historical Case:** In 1799, doctors drained half of **George Washington's** blood to treat a sore throat. He died the next day.
- **The Question:** Did the treatment kill him, or would he have died anyway?
- **The Struggle:** For 2,000 years, doctors believed bloodletting worked because they didn't have a way to prove otherwise.

2. The Fundamental Problem

- **Actual World:**
  - What we see (e.g., Washington was bled and died).
- **Alternative World (Counterfactual):**
  - What would have happened if he wasn't bled.
- **The Gap:**
  - We can never "peek" into that alternative world for a single person.
- **Causal Inference** is the elegant solution to this "insoluble" puzzle

3. The Power of Randomization

- **The PALM Trial (Ebola):** Researchers used a "high-tech coin flip"
  - to assign treatments
- **Fairness in Uncertainty:** When you don't know which drug is better,
  - randomization is the fairest path
- **The Statistical Miracle:**
  - A coin flip balances **seen factors** (age, gender)
    - AND **unseen factors** (genetics, immune system)

4. From Gambler to Casino

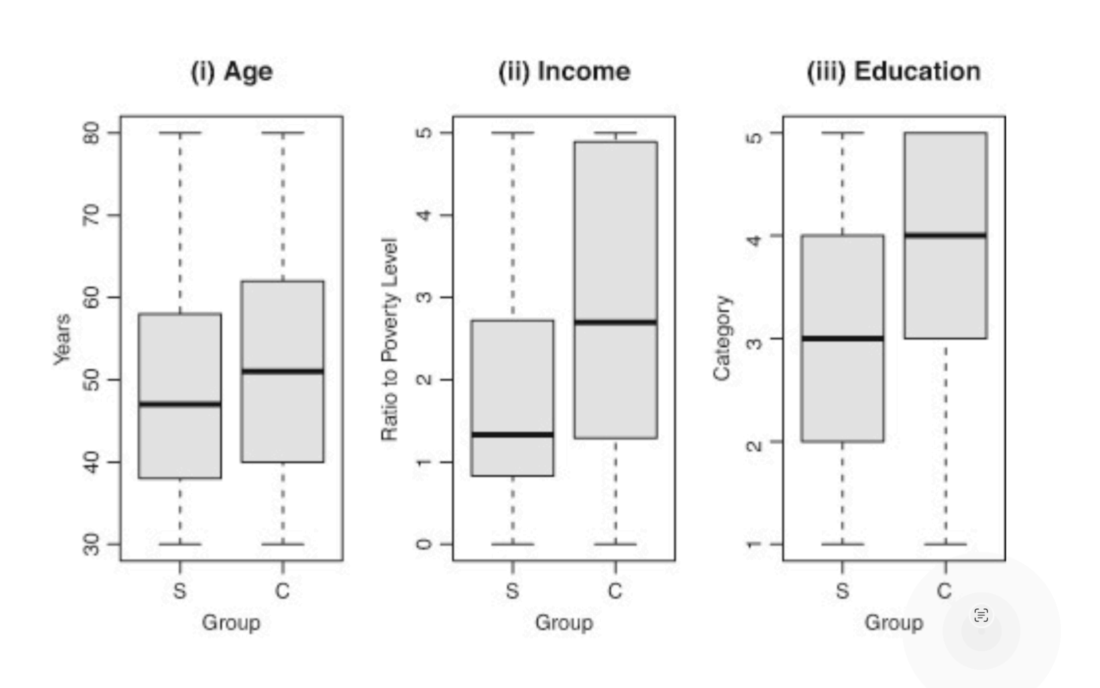
- **Single Flip:**
  - A gamble with no predictable outcome.
- **Hundreds of Flips:**
  - You become the **"casino."**
- **Key Insight:**
  - Over time, randomness cancels out, and the **true average effect** reveals itself
- **Result:** This probability theory, not a biological discovery,
  - solved a life-or-death medical problem

5. When Experiments are Impossible

- **The Ethical Barrier:**
  - We cannot force people to smoke for 40 years to study its effects
- **Observational Data:**
  - We must study the choices people make for themselves.
- **The Problem:**
  - Smokers and non-smokers are different from the start in **age, income, and education.**
- **The Trap:**
  - You aren't comparing "apples to apples."

The Trap: Not comparing "apples to apples"

- Comparison of daily smokers (S) and controls (C)



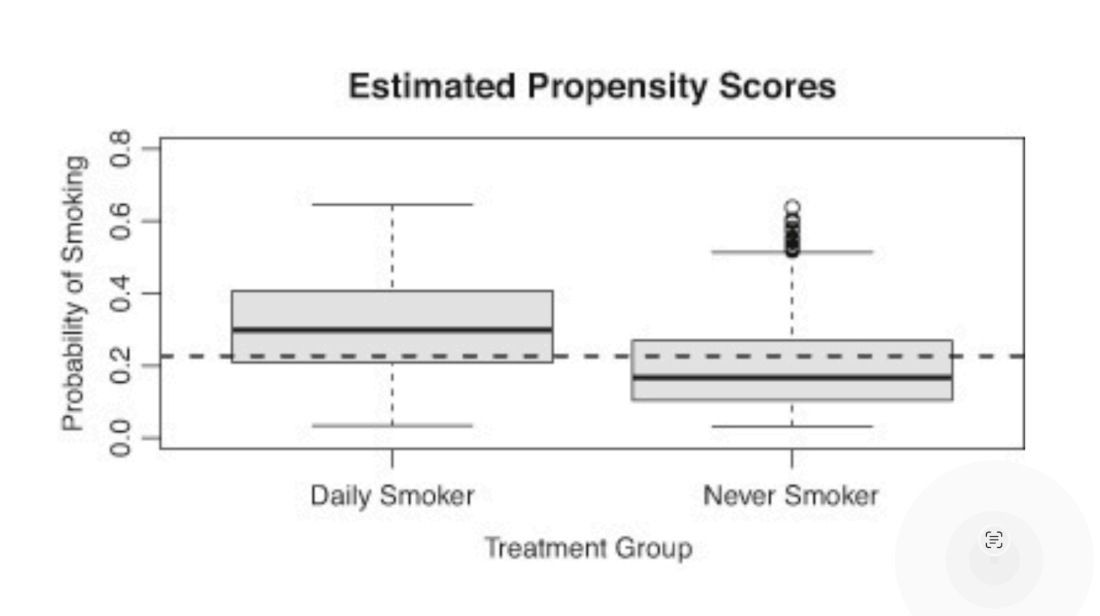
> *credit: "Casual Inference" by Paul Rosenbaum*

6. The Propensity Score

- **Definition:**
  - A single number summarizing the **likelihood** a person
    - would choose a treatment (like smoking) based on their traits
- **The Gap:**
  - Plotting these scores shows that treated and control groups are
    - often not comparable at all
- **Leveling the Field:**
  - We need a way to force this "lopsided" data into a fair comparison

The Gap: Treated and Control not comparable

- Estimated probabilities of smoking (Propensity Score)



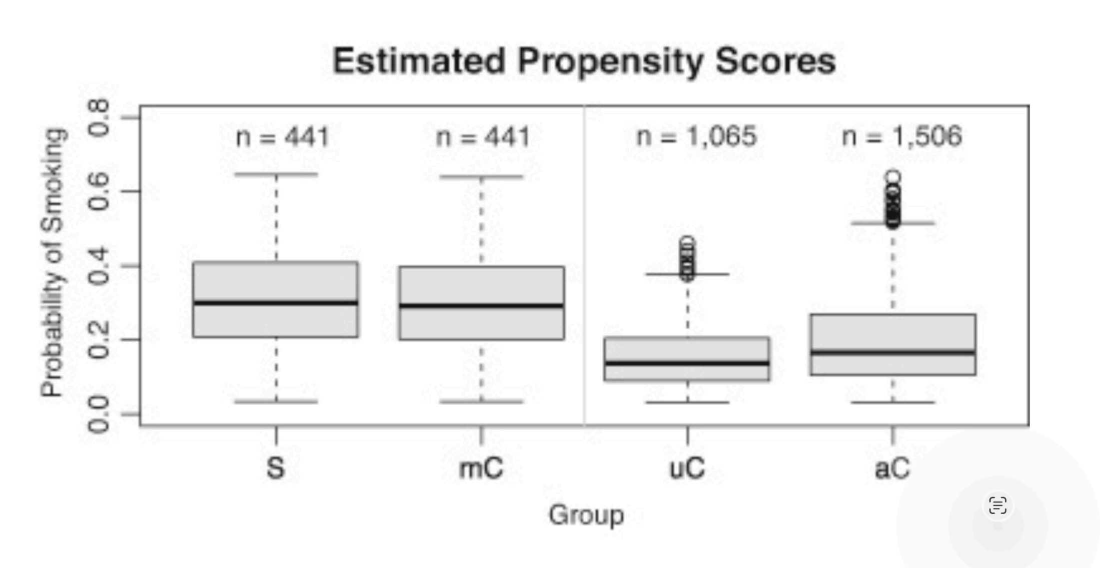
> *credit: "Casual Inference" by Paul Rosenbaum*

7. Statistical Surgery: Matching

- **Statistical Twins:**
  - Matching finds a smoker and a non-smoker
    - with the **exact same probability** of being a smoker
- **Trimming the Noise:**
  - We "toss out" anyone who doesn't have a good match
- **The Outcome:**
  - After matching, the groups (box plots) for age and education look nearly identical

Matching

- S: Smokers, mC: matched Controls,
- uC: unmatched Controls, aC: all Controls



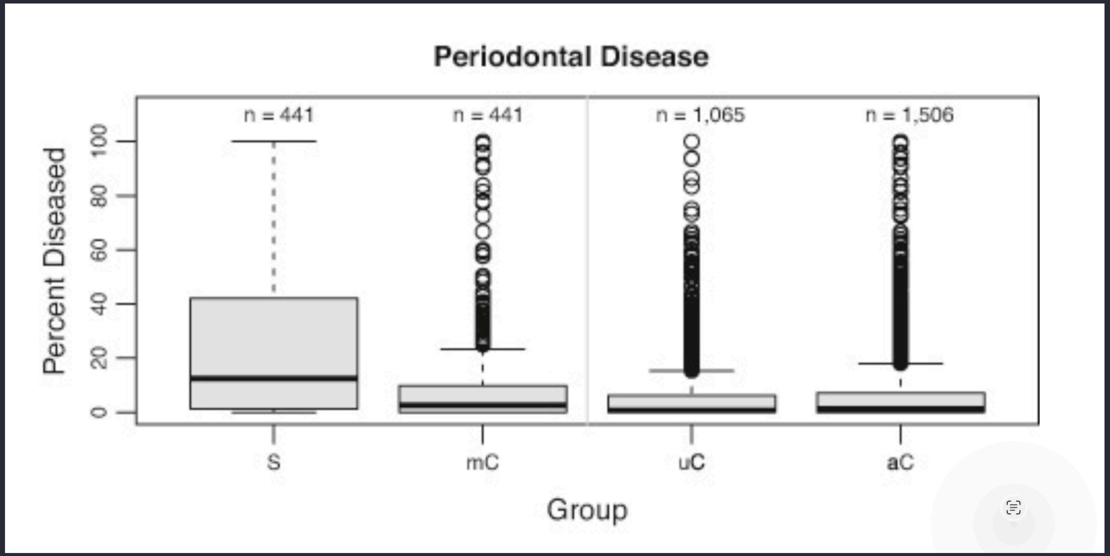
> *credit: "Casual Inference" by Paul Rosenbaum*

8. "Coin People" vs. "Dice People"

- **Coin People:**
  - Individuals who had a high (50/50) chance of treatment
- **Dice People:**
  - Individuals who had a low (1 in 6) chance of treatment
- **Matching's Logic:**
  - It ensures we only compare **coin people to other coin people.**
- **The Result:**
  - Within these matched groups, it is "pretty much random" who ended up treated

Outcome after matching

- Extent of Periodontal disease in smokers and match controls



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9. Turning Noise into Signal

- **The Goal:**
  - Force biased, real-world data to act like a **clean, randomized experiment.**
- **Certainty:**
  - When the gap in outcomes (like periodontal disease) remains after matching,
    - we can be much more certain it is a **real cause**
- **Summary:**
  - We now have the tools to separate genuine cause from mere coincidence

References

Topic	Source
Rosenbaum, P. R. (2023)	Rosenbaum, P. R. (2023). *Causal Inference*. MIT Press.
Henao-Restrepo et al. (2019)	Henao-Restrepo, A. M., et al. (2019). Efficacy and effectiveness of an rVSV-vectored vaccine in preventing Ebola virus disease: final results from the Guinea ring vaccination trial (PALM trial). *New England Journal of Medicine*.